Luca Mazzotta

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Education

York University Toronto, ON

B.Sc., Specialized Honours, Computer Science – Lassonde School of Engineering *Sept. 2023 – Present*

- GPA: 7.57/9.00 | **Relevant Coursework:** Theory of Computation, Computer Organization, Advanced Object-Oriented Programming, Discrete Math, Applied Linear Algebra, Fundamentals of Data Structures, Introduction to AI and Logic Programming, Operating System Fundamentals
- Award of Excellence, Computer Technology (2021–2022) | Honour Roll, Grades 9–12

Experience

Advanced Math Tutor Nov. 2024 – Present

Scholars Education *Bolton, ON*

- Delivered one-on-one and small-group instruction in math and problem-solving to 30+ students across Grades 1–12, breaking down complex concepts into clear, step-by-step explanations
- Diagnosed individual knowledge gaps and adapted lesson plans on the fly, applying a structured, debugging-style approach to isolating and resolving misunderstandings
- Tracked student progress across sessions and iterated on teaching strategies based on measurable outcomes, mirroring an agile, feedback-driven approach to improvement
- Communicated technical and abstract material to non-technical audiences (students and parents), translating complex ideas into accessible language

Camp Counsellor June 2025 – Aug. 2025

Town of Caledon *Bolton, ON*

- Supervised and led daily programming for 20+ campers, planning and facilitating hands-on activities while adapting quickly to the needs of a diverse group
- Coordinated with staff and supervisors to organize materials and logistics for 5+ daily activities, maintaining safety protocols and smooth operations

Projects

Blockify | *Java, Spring Boot, Maven, REST API, Azure* 2025 – Present

- Built a full-stack image processing application that converts any uploaded image into Minecraft-style block art using custom pixel quantization and Euclidean RGB color-matching algorithms
- Designed and implemented a RESTful API with Spring Boot to handle image uploads, processing, and generated output delivery
- Engineered a texture-caching layer with Java HashMaps to reduce redundant color computations and cut image processing latency
- Configured CORS and Spring Security to support a decoupled frontend, and deployed the backend to Azure for scalable, production-ready hosting

Minute Math | *React, Vite, Firebase, JavaScript, Netlify* 2025 – Present

- Developed a daily math puzzle game inspired by Wordle, delivering a new deterministic challenge to all users each day
- Built a custom API to generate and serve daily challenges, ensuring consistent, reproducible puzzles across users and time zones
- Integrated Firebase for data storage and laid groundwork for future user authentication and score persistence
- Designed a clean, responsive UI in React with Vite, deploying continuous updates via Netlify's CI/CD pipeline

Technical Skills

Languages: Java, JavaScript, TypeScript, Python, C, SQL, Swift, HTML/CSS
Frameworks: React, React Native, Node.js, Express.js, Spring Boot, Vite, Tailwind CSS
Libraries: Redux Toolkit, Framer Motion, JUnit, Firebase, MathQuill,
Developer Tools: Git, GitHub, Docker, VS Code, IntelliJ IDEA, Eclipse, Visual Studio, Postman, Figma, Microsoft Azure, Netlify, Render, Cloudflare